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16: Continuous Joint Distributions

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♥ February 14, 2024 ♥

[Lecture Discussion on Ed](#)

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Continuous Joint Distributions

Stanford logo with darts



The Stanford letterhead logo was created by throwing 500,000 darts according to a joint distribution.

not really, but let's pretend

If we throw another dart according to the same distribution, what is

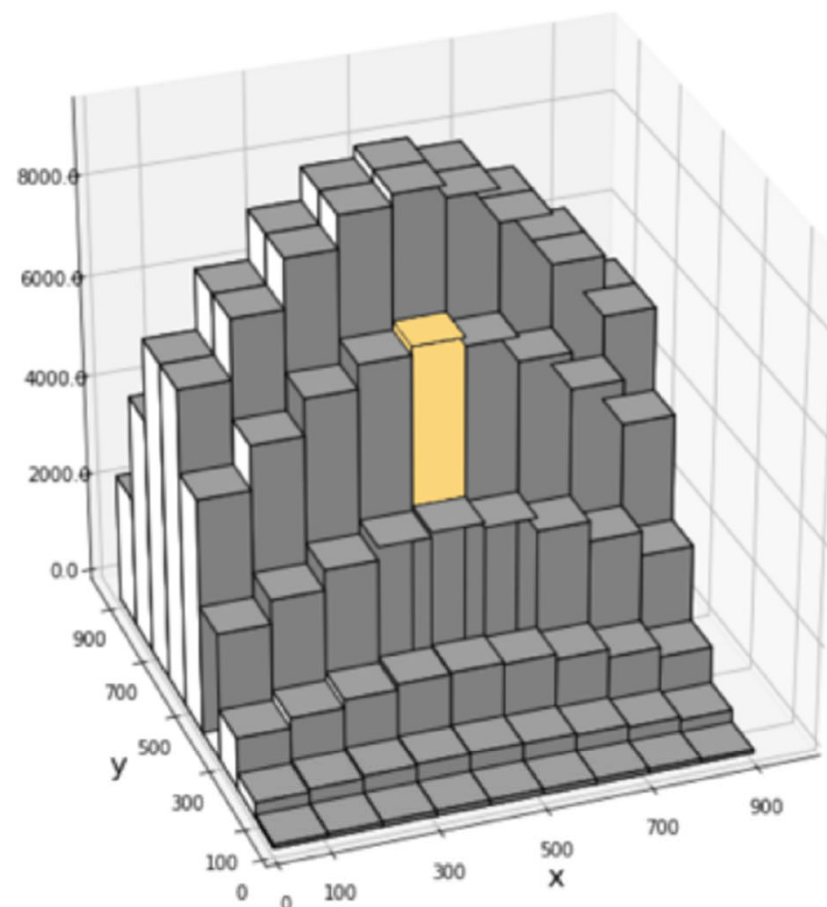
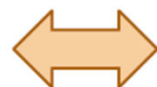
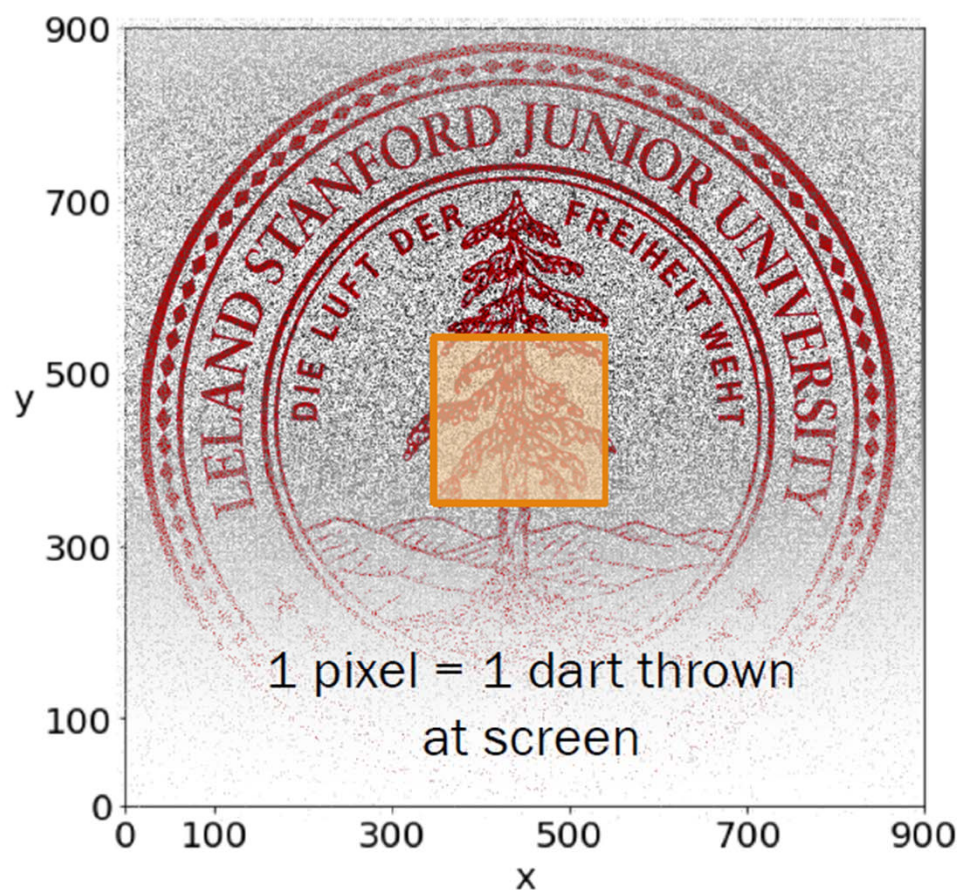
$P(\text{dart hits within } r \text{ pixels of center})?$

Quick check: What is the probability that a dart hits at (456.2344132343, 532.1865739012)?

so small it's essentially zero

CS109 logo with darts

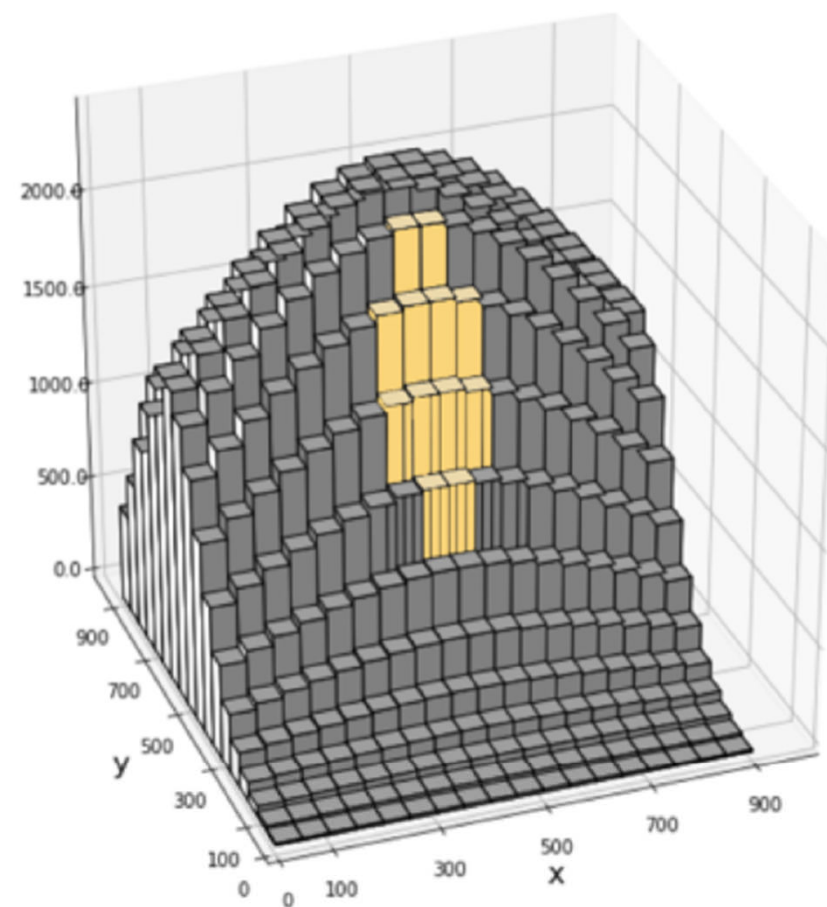
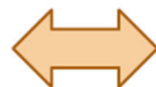
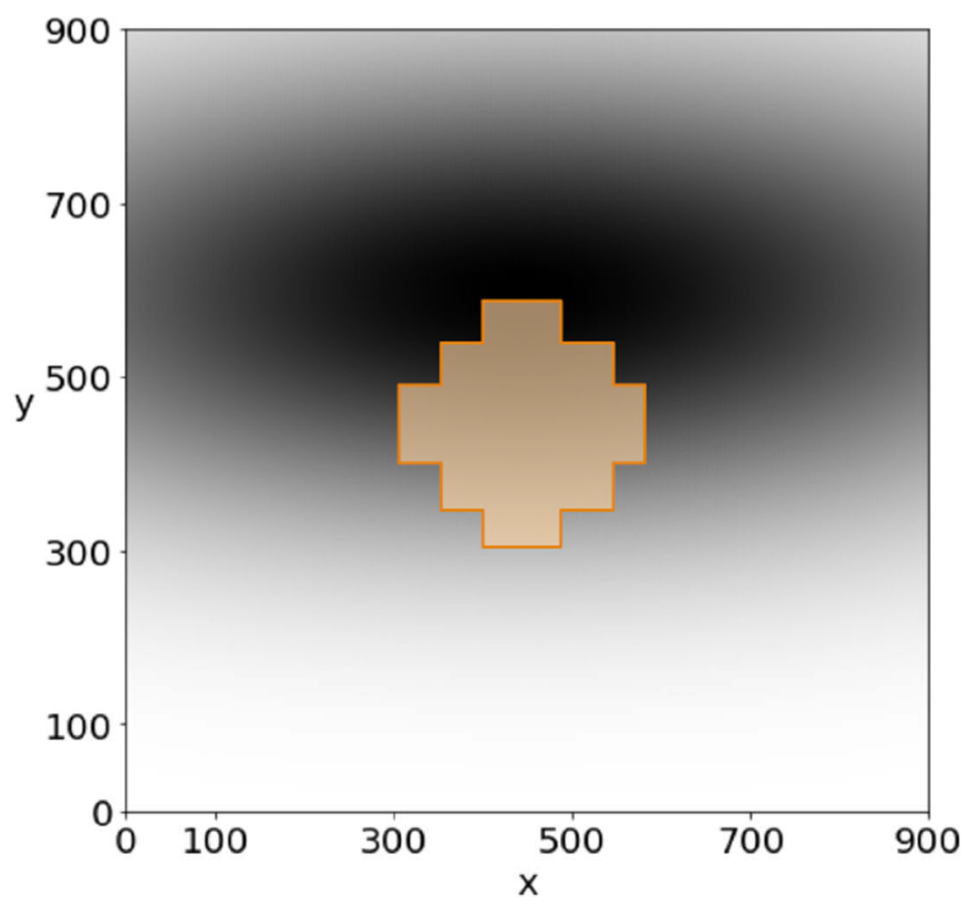
$P(\text{dart hits within } r \text{ pixels of center})?$



Possible dart counts (in 100x100 boxes)

CS109 logo with darts

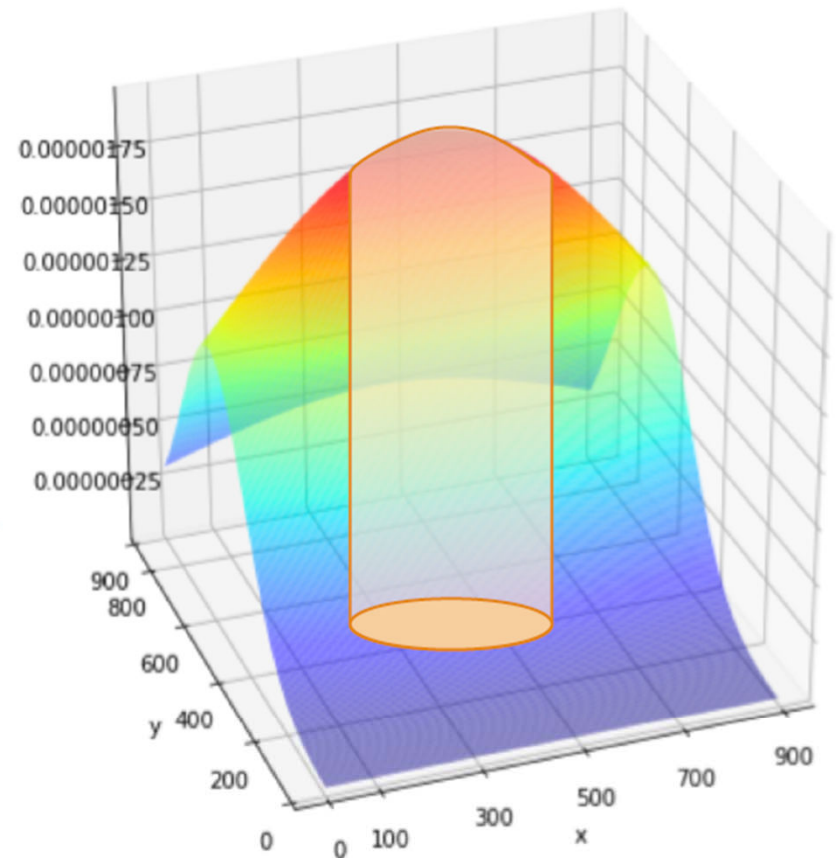
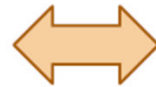
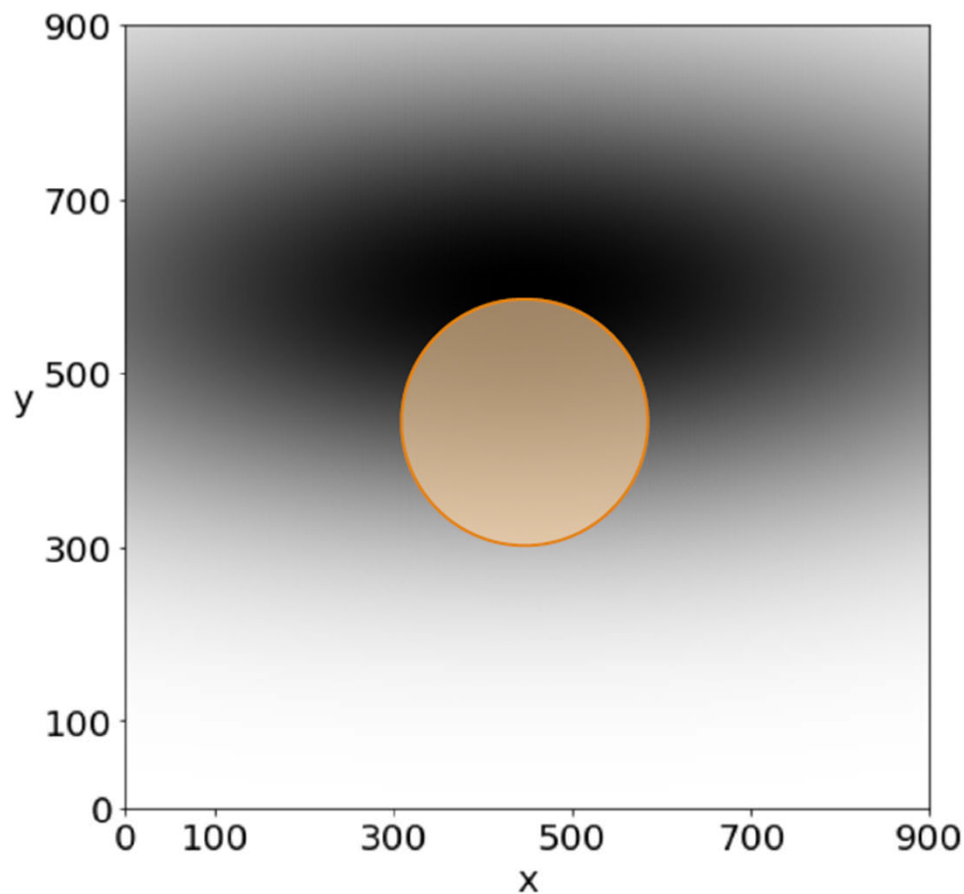
$P(\text{dart hits within } r \text{ pixels of center})?$



Possible dart counts (in 50x50 boxes)

CS109 logo with darts

$P(\text{dart hits within } r \text{ pixels of center})?$



Possible dart counts
(in infinitesimally small boxes) iversity

Continuous joint probability density functions

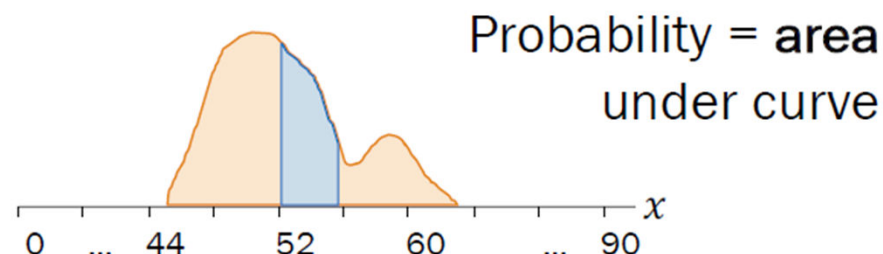
If two random variables X and Y are jointly continuous, then there exists a **joint probability density function** $f_{X,Y}$ defined over $-\infty < x, y < \infty$ such that:

$$P(a_1 \leq X \leq a_2, b_1 \leq Y \leq b_2) = \int_{a_1}^{a_2} \int_{b_1}^{b_2} f_{X,Y}(x, y) dy dx$$

From one continuous RV to jointly continuous RVs

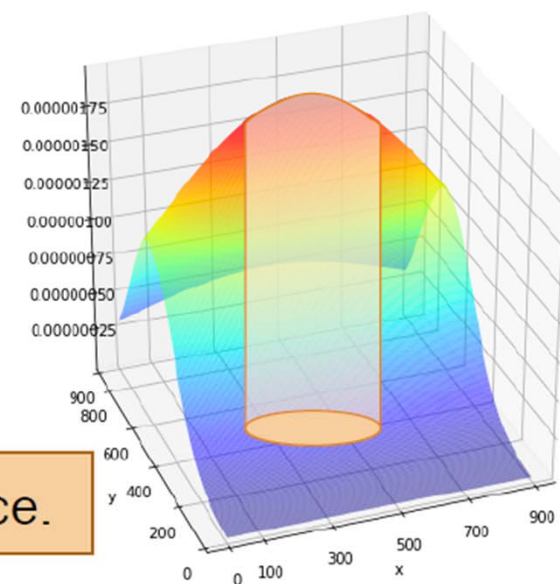
Single continuous RV X

- PDF f_X such that $\int_{-\infty}^{\infty} f_X(x) dx = 1$
- Integrate to get probabilities



Jointly continuous RVs X and Y

- PDF $f_{X,Y}$ such that $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy dx = 1$
- Double integrate to get probabilities



Probability for jointly continuous RVs is **volume** under a surface.

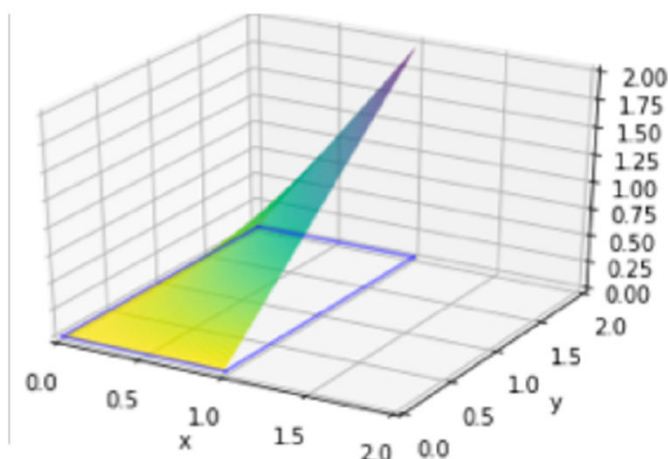
Double integrals without tears

Let X and Y be two continuous random variables.

- Support: $0 \leq X \leq 1$, $0 \leq Y \leq 2$.

Is $g(x, y) = xy$ a valid joint PDF over X and Y ?

Write down the definite double integral that must integrate to 1:



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Double integrals without tears

Let X and Y be two continuous random variables.

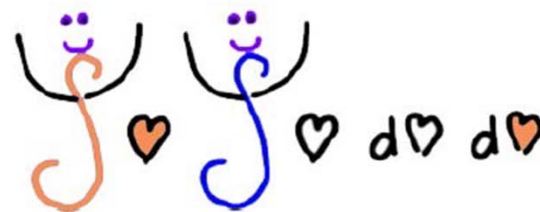
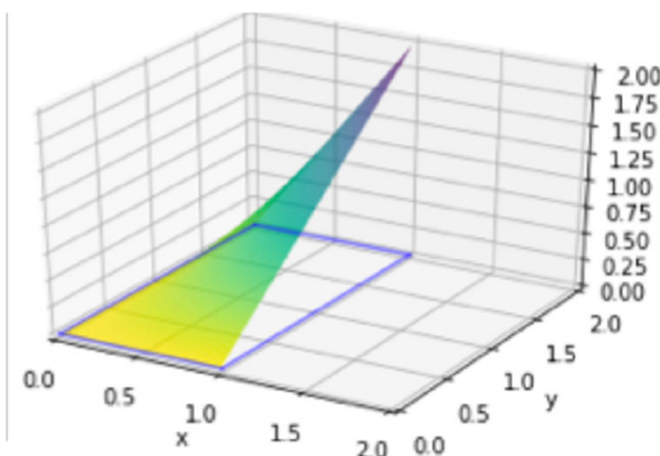
- Support: $0 \leq X \leq 1, 0 \leq Y \leq 2$.

Is $g(x, y) = xy$ a valid joint PDF over X and Y ?

Write down the definite double integral that must integrate to 1:

$$\int_{y=0}^2 \int_{x=0}^1 xy \, dx \, dy = 1 \quad \text{or} \quad \int_{x=0}^1 \int_{y=0}^2 xy \, dy \, dx = 1$$

(used in next slide)



Double integrals without tears

Let X and Y be two continuous random variables.

- Support: $0 \leq X \leq 1, 0 \leq Y \leq 2$.

Is $g(x, y) = xy$ a valid joint PDF over X and Y ?

0. Set up integral:

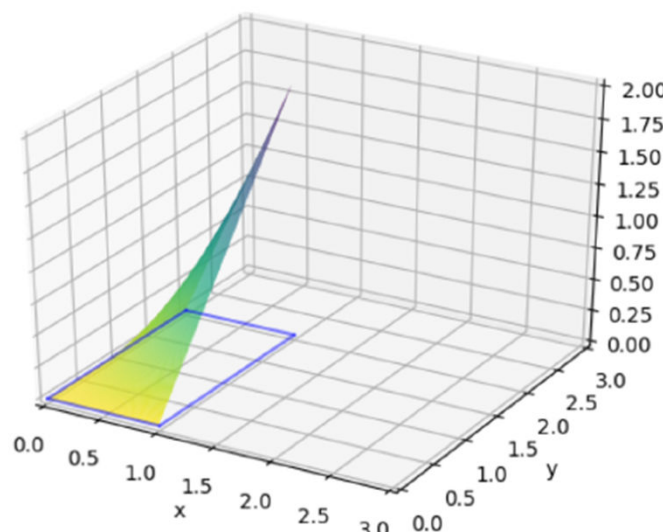
$$1 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) dx dy = \int_{y=0}^2 \int_{x=0}^1 xy dx dy$$

1. Evaluate inside integral by treating y as a constant:

$$\int_{y=0}^2 \left(\int_{x=0}^1 xy dx \right) dy = \int_{y=0}^2 y \left(\int_{x=0}^1 x dx \right) dy = \int_{y=0}^2 y \left[\frac{x^2}{2} \right]_0^1 dy = \int_{y=0}^2 y \frac{1}{2} dy$$

2. Evaluate remaining (single) integral:

$$\int_{y=0}^2 y \frac{1}{2} dy = \left[\frac{y^2}{4} \right]_{y=0}^2 = 1 - 0 = 1$$

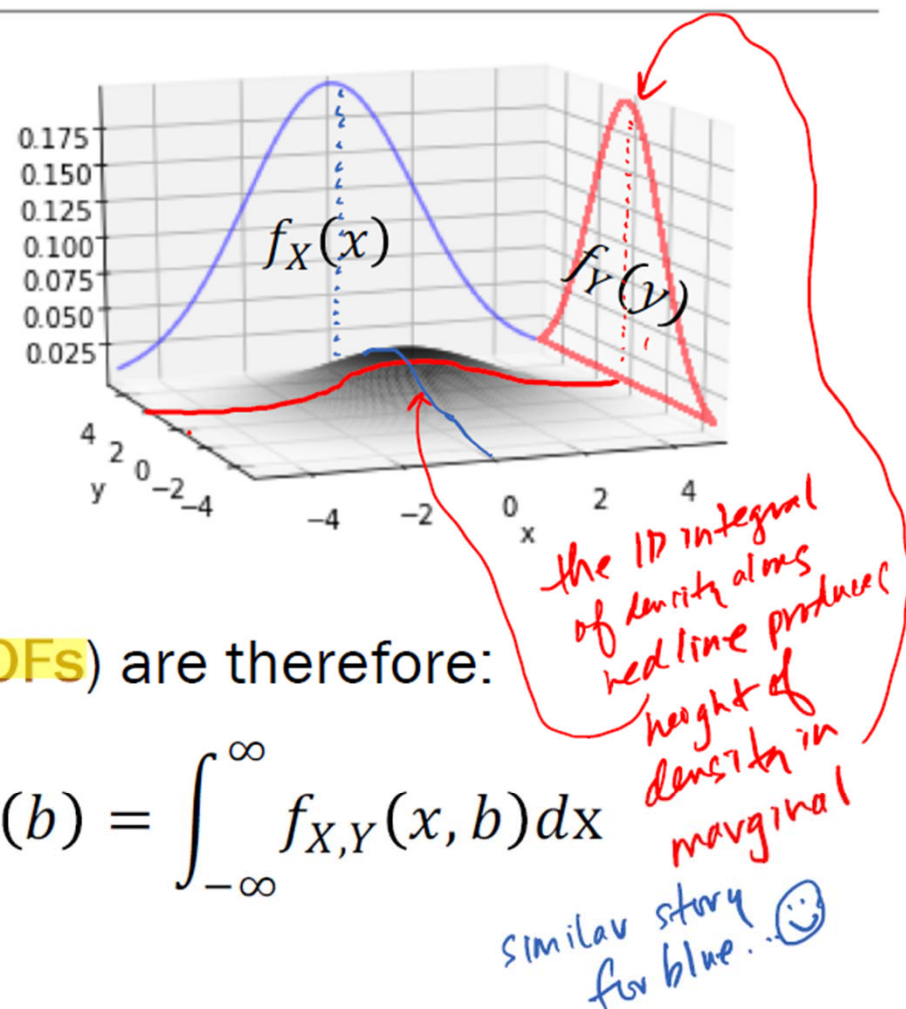


Yes, $g(x, y)$ is a valid joint PDF because it integrates to 1.

Marginal distributions

Suppose X and Y are continuous random variables with joint PDF:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy dx = 1$$



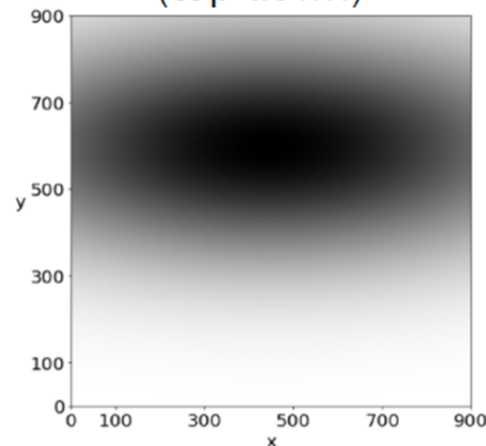
The marginal density functions (**marginal PDFs**) are therefore:

$$f_X(a) = \int_{-\infty}^{\infty} f_{X,Y}(a,y) dy$$

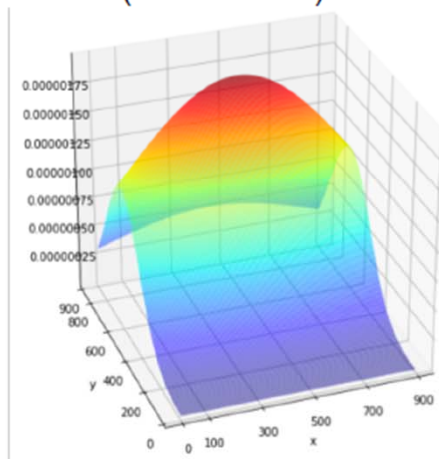
$$f_Y(b) = \int_{-\infty}^{\infty} f_{X,Y}(x,b) dx$$

Back to darts!

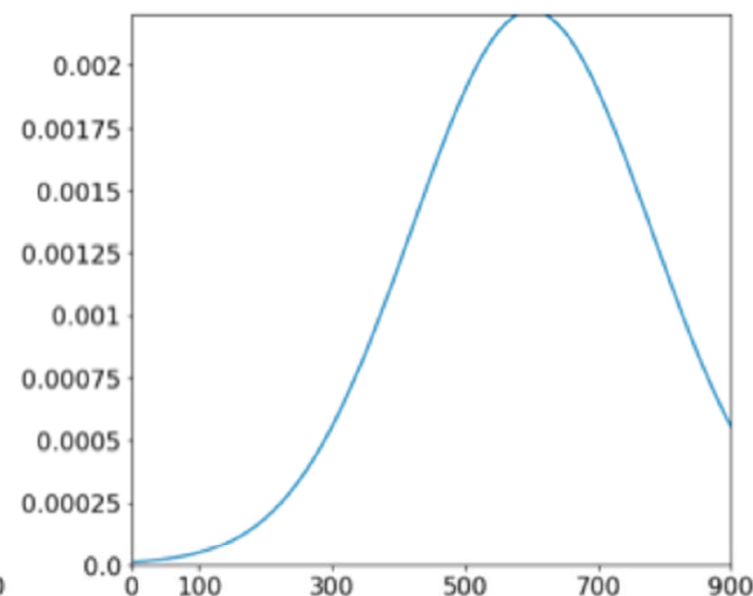
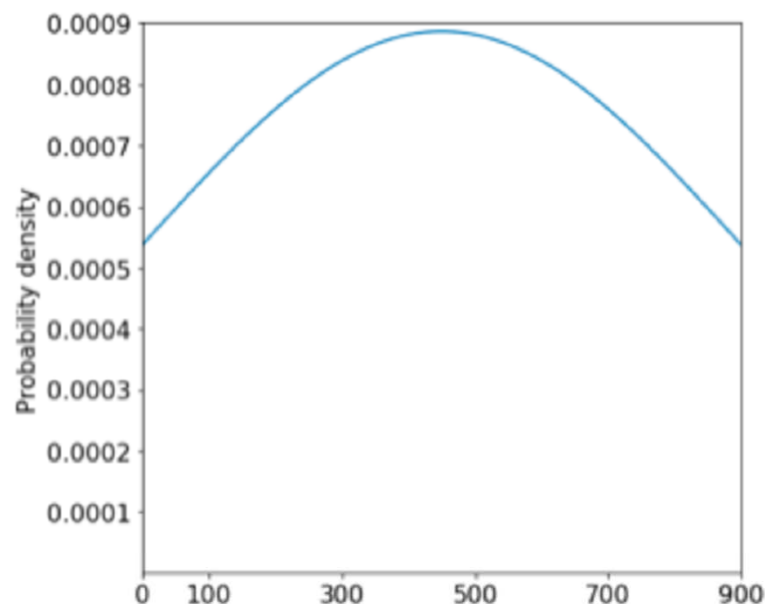
(top-down)



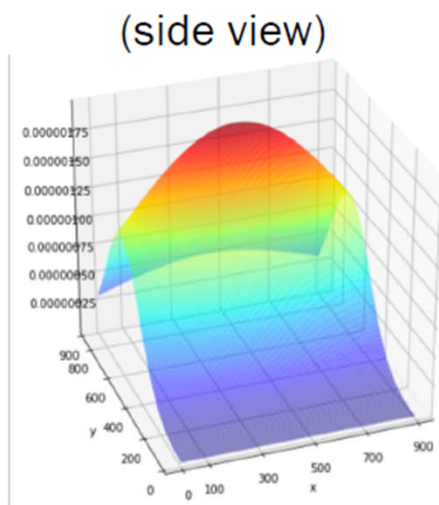
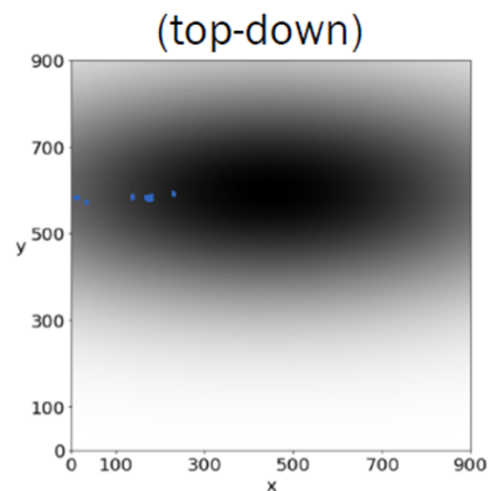
(side view)



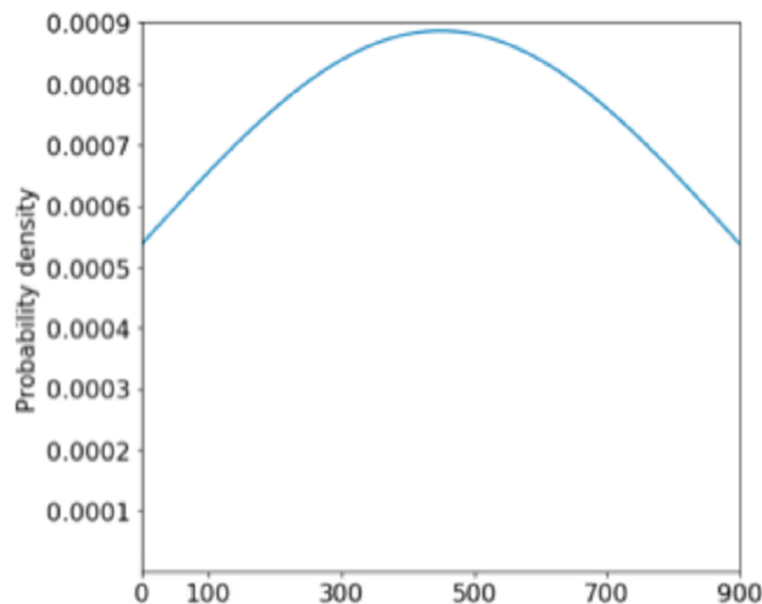
Match X and Y to their respective marginal PDFs:



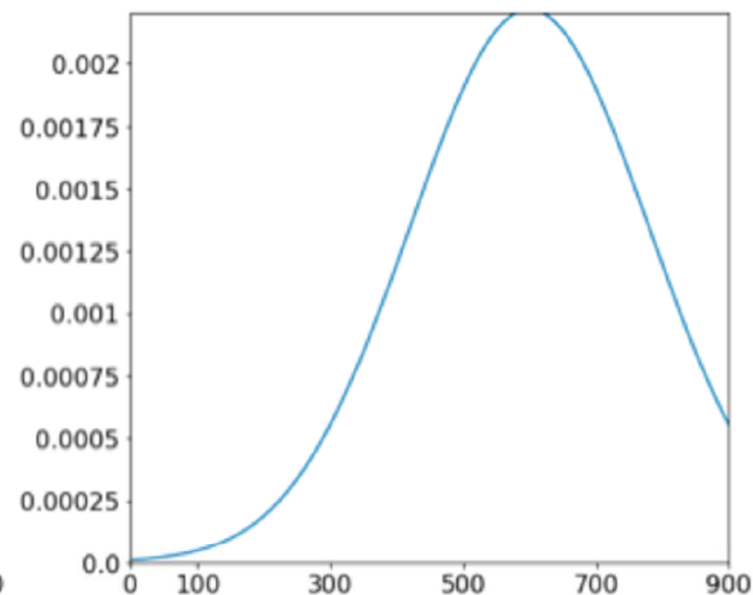
Back to darts!



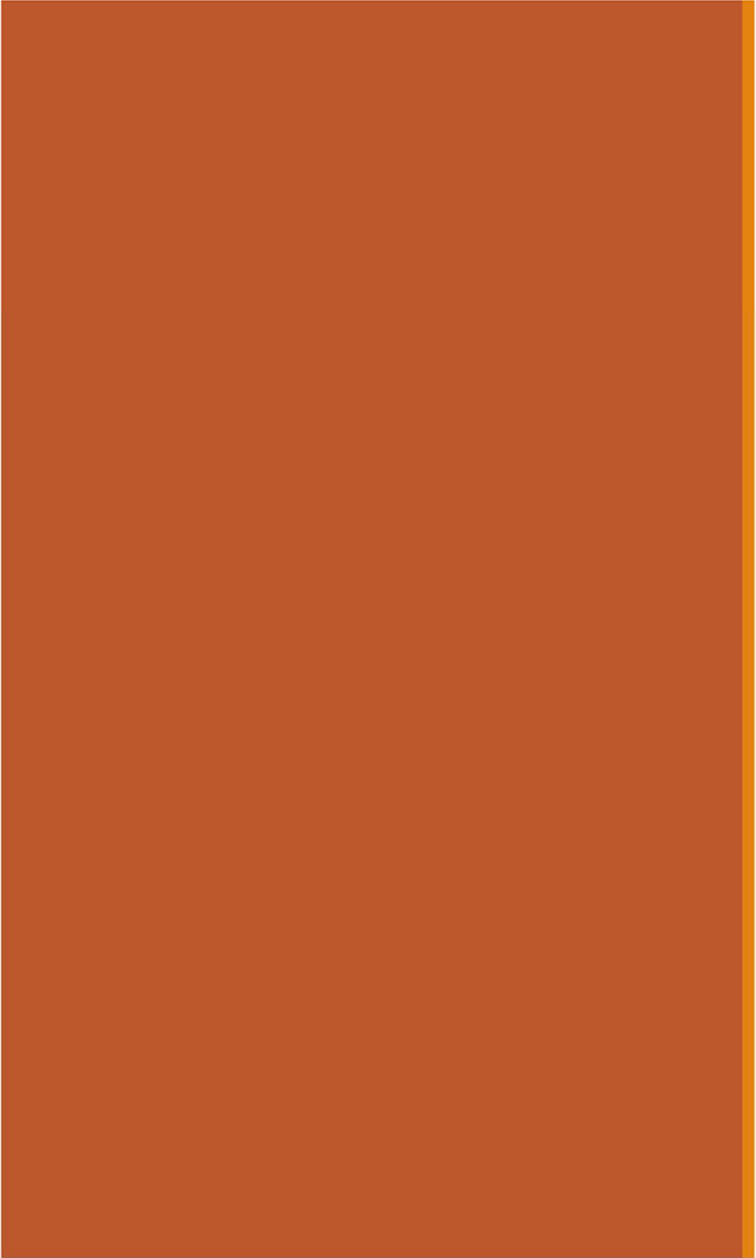
Match X and Y to their respective marginal PDFs:



pixel x



pixel y

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Joint CDFs

An observation: Connecting CDF to PDF

For a continuous random variable X with PDF f , the CDF (cumulative distribution function) is

$$F(a) = P(X \leq a) = \int_{-\infty}^a f(x)dx$$

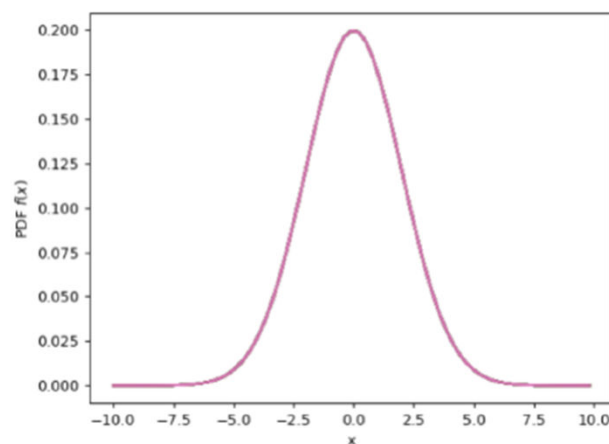
The density f is therefore the derivative of the CDF, F :

$$f(x) = \frac{d}{dx}F(x)$$

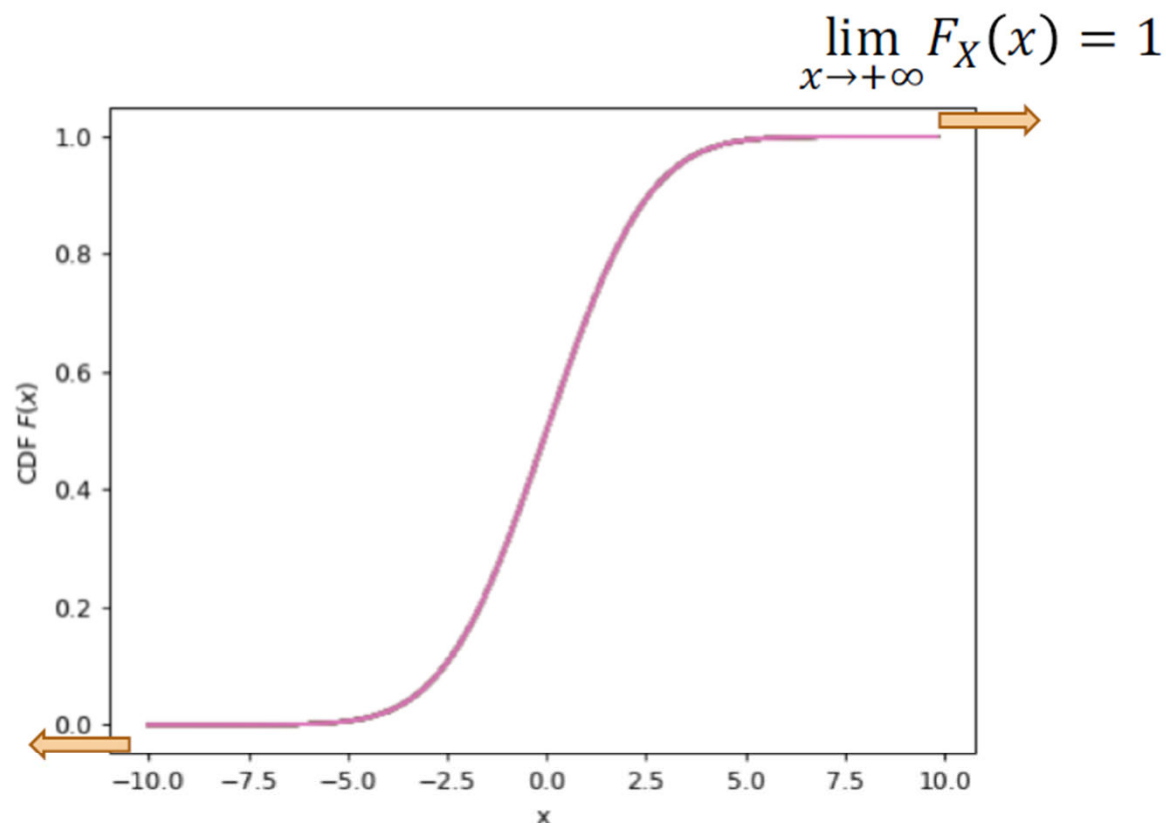
Fundamental Theorem of
Calculus

Single variable CDF, graphically

Review



$f_X(x)$



$$\lim_{x \rightarrow -\infty} F_X(x) = 0$$

$$\lim_{x \rightarrow +\infty} F_X(x) = 1$$

$$F_X(x) = P(X \leq x)$$

Joint cumulative distribution function

For two random variables X and Y , there can be a **joint cumulative distribution function** $F_{X,Y}$:

$$F_{X,Y}(a, b) = P(X \leq a, Y \leq b)$$

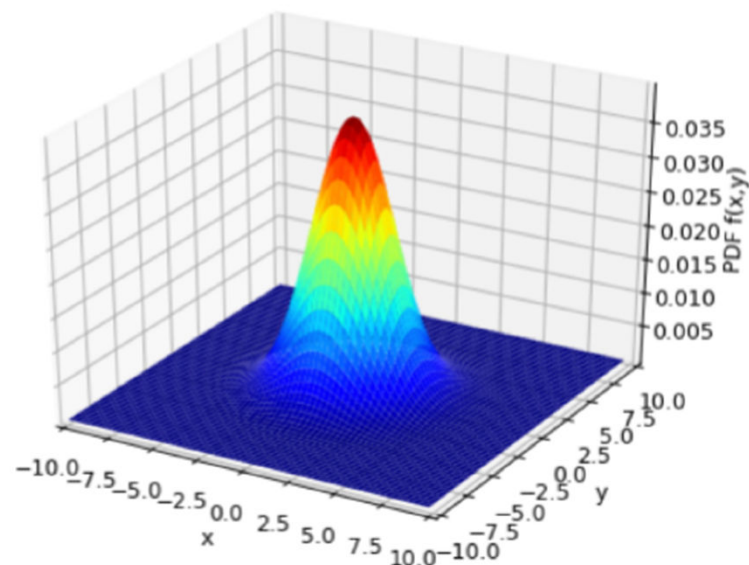
For discrete X and Y :

$$F_{X,Y}(a, b) = \sum_{x \leq a} \sum_{y \leq b} p_{X,Y}(x, y)$$

For continuous X and Y :

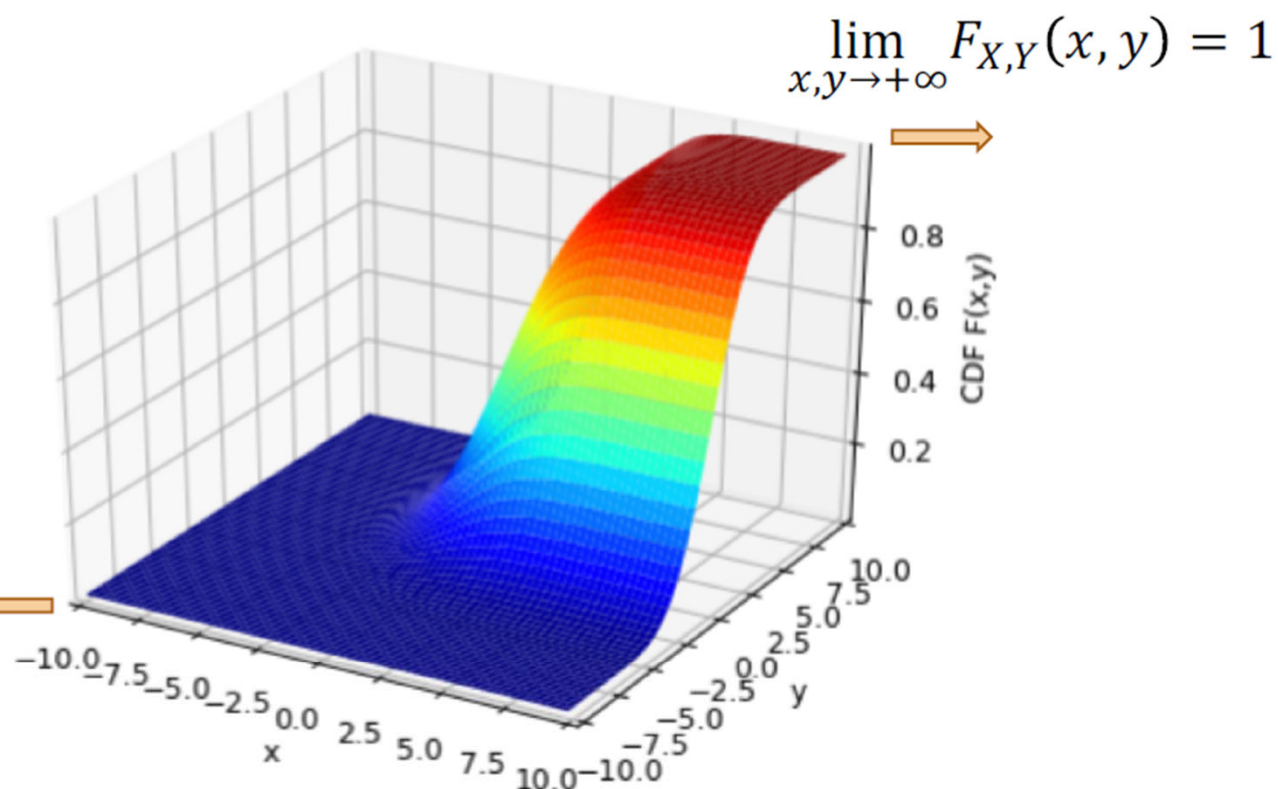
$$F_{X,Y}(a, b) = \int_{-\infty}^a \int_{-\infty}^b f_{X,Y}(x, y) dy dx$$
$$f_{X,Y}(a, b) = \frac{\partial^2}{\partial a \partial b} F_{X,Y}(a, b)$$

Joint CDF, graphically

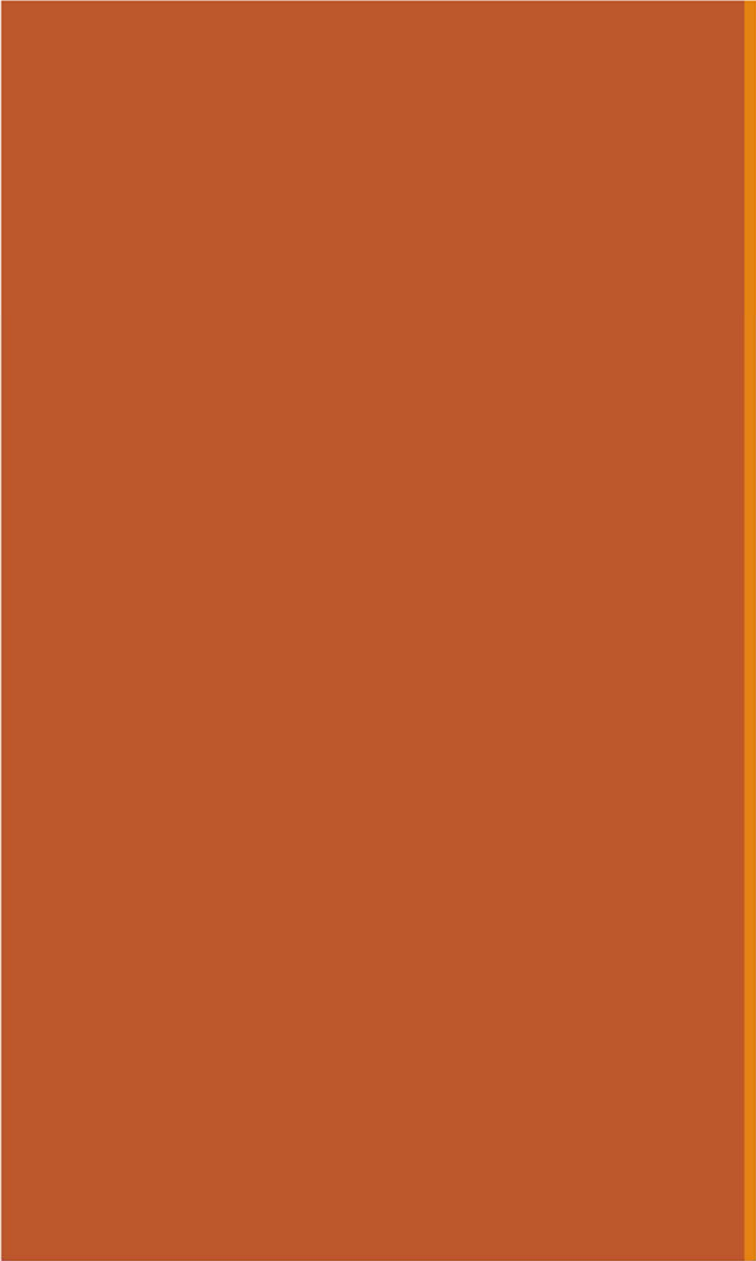


$$\lim_{x,y \rightarrow -\infty} F_{X,Y}(x,y) = 0$$

$$f_{X,Y}(x,y)$$



$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$$

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Independent Continuous RVs

Independent continuous RVs

Two continuous random variables X and Y are **independent** if:

$$P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y) \quad \forall x, y$$

Equivalently:

$$\begin{aligned} F_{X,Y}(x, y) &= F_X(x)F_Y(y) \\ f_{X,Y}(x, y) &= f_X(x)f_Y(y) \end{aligned} \quad \forall x, y$$

Proof of PDF:

$$\begin{aligned} f_{X,Y}(x, y) &= \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y) = \frac{\partial^2}{\partial x \partial y} F_X(x)F_Y(y) \\ &= \frac{\partial}{\partial x} \frac{\partial}{\partial y} \underline{F_X(x)F_Y(y)} = \frac{\partial}{\partial x} F_X(x) \frac{\partial}{\partial y} F_Y(y) \\ &= f_X(x)f_Y(y) \end{aligned}$$

this is a constant with respect to y!

Independent continuous RVs

Two continuous random variables X and Y are independent if:

$$P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y) \quad \forall x, y$$

Equivalently:

$$\begin{aligned} F_{X,Y}(x, y) &= F_X(x)F_Y(y) \\ f_{X,Y}(x, y) &= f_X(x)f_Y(y) \end{aligned} \quad \forall x, y$$

More generally, X and Y are **independent** if joint density on factors into two, single-variable marginal probability densities:

$$f_{X,Y}(x, y) = g(x)h(y), \text{ where } -\infty < \underbrace{x, y}_{\text{states that } x \text{ and } y \text{ can freely roam over their full support!}} < \infty$$

Pop quiz! (more kidding)

$f_{X,Y}(x,y) = g(x)h(y)$,
where $-\infty < x, y < \infty$  independent
X and Y

Are X and Y independent in the following cases?

✓ 1. $f_{X,Y}(x,y) = 6e^{-3x}e^{-2y}$
where $0 < x, y < \infty$

Separable functions: $g(x) = 3e^{-3x}$

$$h(y) = 2e^{-2y}$$

$$\int_0^1 \int_0^1 4xy = 1 = \int_0^1 C 4x dx \cdot \int_0^1 \frac{1}{C} y dy$$

Separable functions: $g(x) = 2x$

$$h(y) = 2y$$

Constant is
dictated
by requirement
that yellow
integrate to 1 and
blue integrate to 1.
 $C = 1/2$

✓ 2. $f_{X,Y}(x,y) = 4xy$
where $0 < x, y < 1$

3. $f_{X,Y}(x,y) = 24xy$
where $0 < x + y < 1$ ✓

Cannot capture constraint on $x + y$!

If you can factor densities over the entire support, you have independence.

More pop quiz! (more kidding)

X and Y have the following joint PDF:

$$f_{X,Y}(x,y) = 3e^{-3x}$$

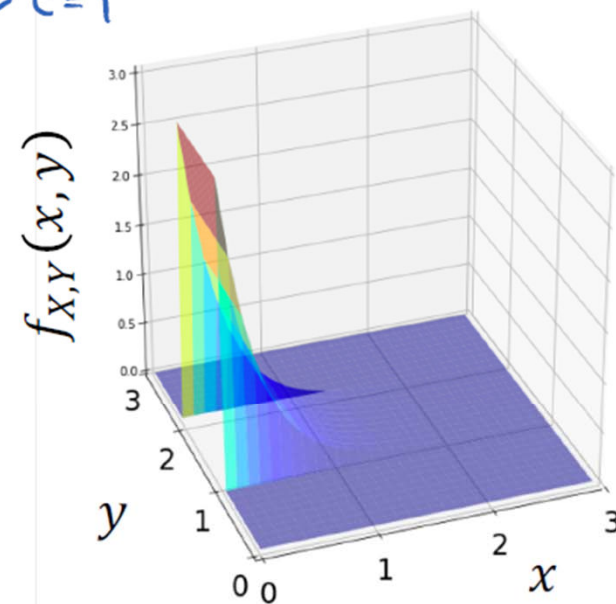
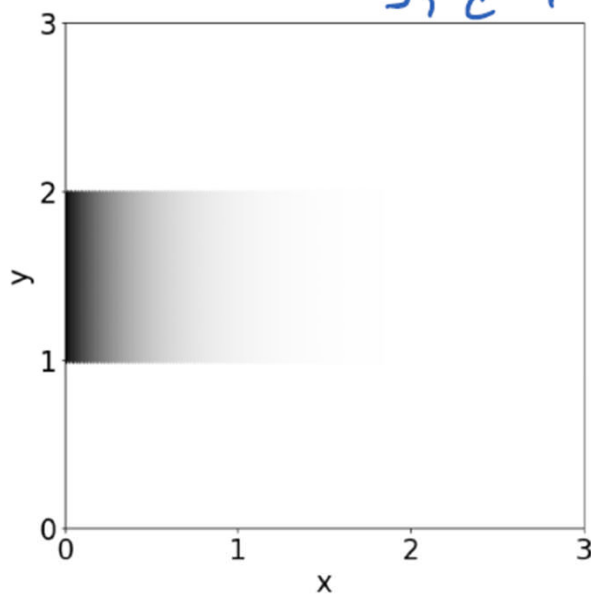
where $0 < x < \infty, 1 < y < 2$

1. Are X and Y independent?

$$g(x) = 3Ce^{-3x}, 0 < x < \infty \quad C \text{ is a}$$
$$h(y) = 1/C, \quad 1 < y < 2 \quad \text{constant}$$
$$\int_1^2 \frac{1}{C} dy = 1 \Rightarrow C = 1$$

2. What is the marginal PDF of X ? Of Y ?

3. What is $E[X + Y]$?



More pop quiz! (more kidding)

X and Y have the following joint PDF:

$$f_{X,Y}(x,y) = 3e^{-3x} \\ \text{where } 0 < x < \infty, 1 < y < 2$$

1. Are X and Y independent?

$$g(x) = 3e^{-3x}, 0 < x < \infty \\ h(y) = 1, 1 < y < 2$$

2. What is the marginal PDF of X ? Of Y ?

$$\begin{cases} f_Y(y) = h(y) = 1, 1 < y < 2 \Leftrightarrow Y \sim \text{Uni}(1,2) \\ f_X(x) = g(x) = 3e^{-3x}, 0 < x < \infty \Leftrightarrow X \sim \text{Exp}(3) \end{cases}$$

3. What is $E[X + Y]$?

$$\text{Strategy 1: } E[g(x,y)] = \int \int g(x,y) f_{X,Y}(x,y) dy dx \quad \text{LOTS} \\ = \int \int (x+y) f_{X,Y}(x,y) dy dx$$

$$\text{Strategy 2: } E[X+Y] = E[X] + E[Y] \\ = 1/3 + 3/2 = 11/6$$

Linearity of expectation


The joy of meetings

Two people set up a meeting time. Each arrives independently at a time uniformly distributed between 12pm and 12:30pm.

Define $X = \#$ minutes past 12pm that person 1 arrives. $X \sim \text{Uni}(0, 30)$

$Y = \#$ minutes past 12pm that person 2 arrives. $Y \sim \text{Uni}(0, 30)$

✓ What is the probability that the first to arrive waits > 10 mins for the other?

Compute: $P(X + 10 < Y) + P(Y + 10 < X) = 2P(X + 10 < Y)$ (by symmetry)

1. What is symmetry here?
2. How do we integrate to compute this probability?

$$f_{X,Y}(x,y) = \left(\frac{1}{30}\right)^2$$

$$\underbrace{\iint_{\substack{0 \leq x,y \leq 30 \\ x+10 < y}} \left(\frac{1}{30}\right)^2 dx dy}_{\text{Region 1}} + \underbrace{\iint_{\substack{0 \leq x,y \leq 30 \\ y+10 < x}} \left(\frac{1}{30}\right)^2 dx dy}_{\text{Region 2}}$$

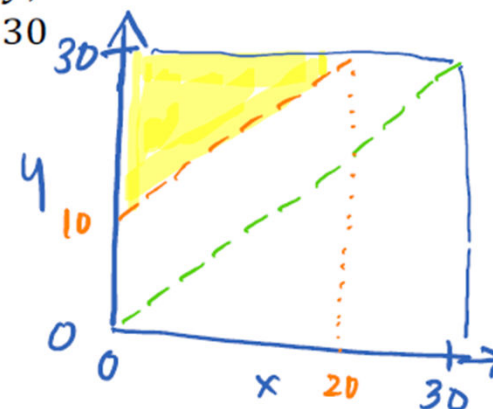
Double integrals: A guide

From last slide: $2P(X + 10 < Y) = 2 \iint_{\substack{x+10 < y, \\ 0 \leq x, y, \leq 30}} (1/30)^2 dx dy$

(by symmetry, independence)

Steps:

1. Draw a picture.
2. Set bounds "from outside in".
 - Outer integral bounds should be full range possible
 - Inner integral can depend on integration variable of outer integral



blue box
delineates
full range of
2D PDF ✓

green dashed
line is
 $y = x$ ✓

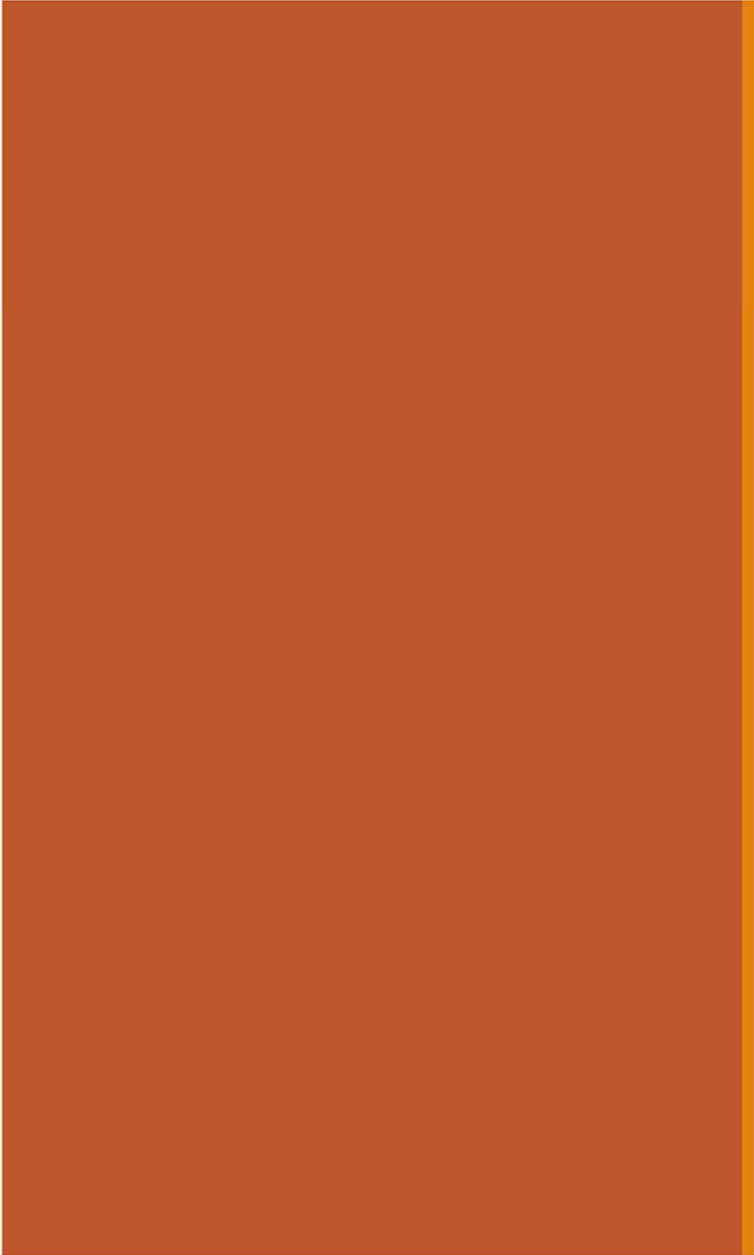
orange dashed
line is
 $y = x + 10$ ✓

yellow is everywhere
 $y > x + 10$ ✓

$$= \frac{2}{30^2} \int_{10}^{30} \int_0^{y-10} dx dy$$

$$= \frac{2}{30^2} \int_{10}^{30} (y - 10) dy$$

$$= \dots = \frac{9}{9}$$

A solid orange horizontal bar spanning the width of the slide, positioned at the top.

Bivariate Normal Distribution

(랜덤변수 2개)

Bivariate Normal Distribution

recall that for $X \sim \mathcal{N}(\mu, \sigma^2) \Rightarrow f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

X_1 and X_2 follow a bivariate normal distribution if their joint PDF f is

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left(\frac{(x_1-\mu_1)^2}{\sigma_1^2} - \underbrace{\frac{2\rho(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2}} + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right)}$$

Can show that $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2), X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$

(Ross chapter 6, example 5d)

Often written as:

$$\underline{\mathbf{X}} \sim \mathcal{N}(\underline{\boldsymbol{\mu}}, \underline{\boldsymbol{\Sigma}})$$

- Vector $\mathbf{X} = (X_1, X_2)$

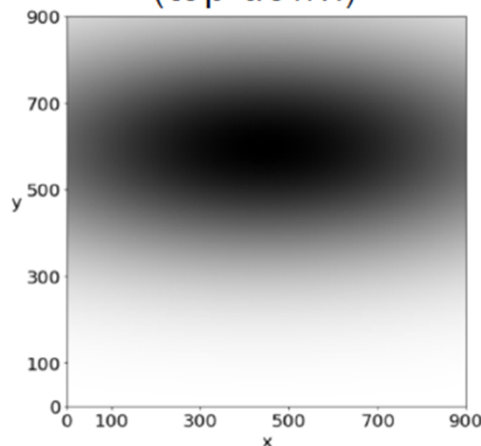
- Mean vector $\boldsymbol{\mu} = (\mu_1, \mu_2)$, Covariance matrix: $\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix} = \begin{bmatrix} \text{Cov}(X_1, X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Cov}(X_2, X_2) \end{bmatrix}$

Recall correlation: $\rho = \frac{\text{Cov}(X_1, X_2)}{\sigma_1\sigma_2}$

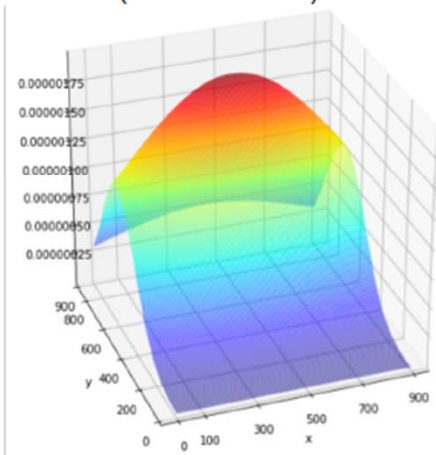
We will focus on understanding the **shape** of a bivariate Normal RV.

Back to darts

(top-down)



(side view)



Darts were thrown according to a bivariate normal distribution:

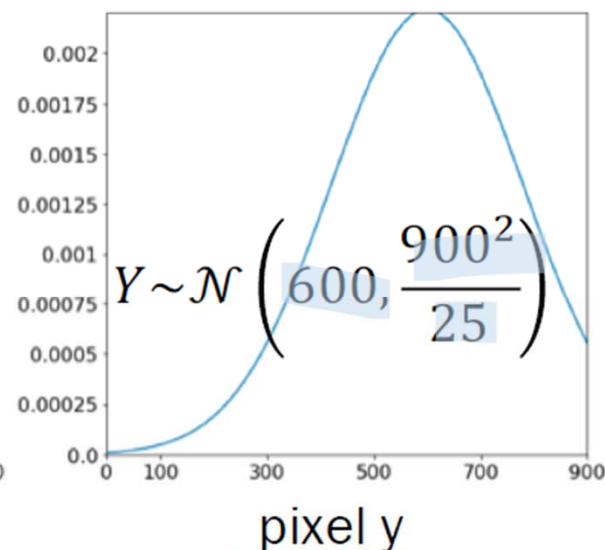
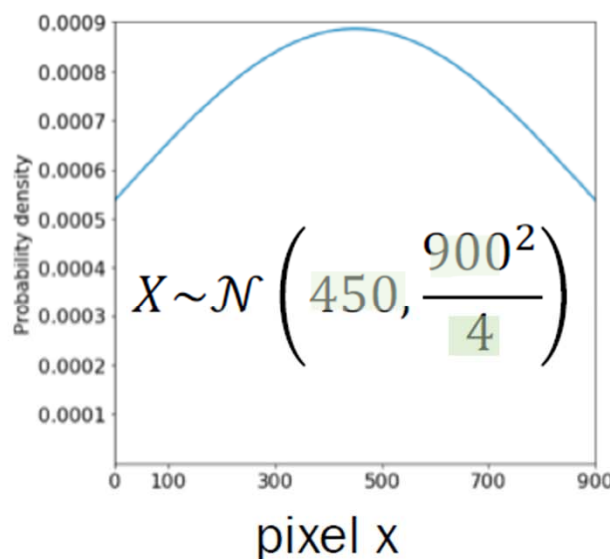
$$(X, Y) \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

$$\boldsymbol{\mu} = (450, 600)$$

$$\boldsymbol{\Sigma} = \begin{bmatrix} 900^2/4 & 0 \\ 0 & 900^2/25 \end{bmatrix}$$

says that
 $\text{Cov}(X, Y) = 0$

Marginal
PDFs:



A diagonal covariance matrix

Let $X = (X_1, X_2)$ follow a bivariate normal distribution $X \sim \mathcal{N}(\mu, \Sigma)$, where

$$\mu = (\mu_1, \mu_2),$$

$$\Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$$

still says $\text{Cov}(X_1, X_2) = 0$

Are X_1 and X_2 independent?

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left(\frac{(x_1-\mu_1)^2}{\sigma_1^2} - \underbrace{\frac{2\rho(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2}}_{\text{goes to zero}} + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right)}$$

$$= \frac{1}{2\pi\sigma_1\sigma_2} e^{-\frac{1}{2}\left(\frac{(x_1-\mu_1)^2}{\sigma_1^2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right)}$$

(Note covariance: $\rho\sigma_1\sigma_2 = 0$)

$$= \frac{1}{\sigma_1\sqrt{2\pi}} e^{-(x_1-\mu_1)^2/2\sigma_1^2} \frac{1}{\sigma_2\sqrt{2\pi}} e^{-(x_2-\mu_2)^2/2\sigma_2^2}$$



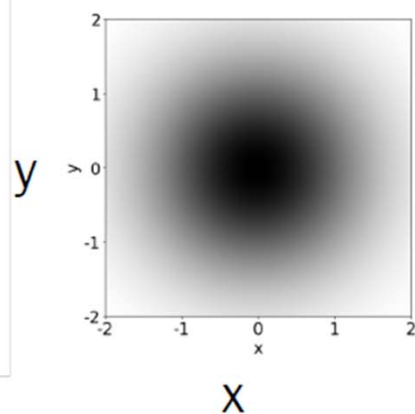
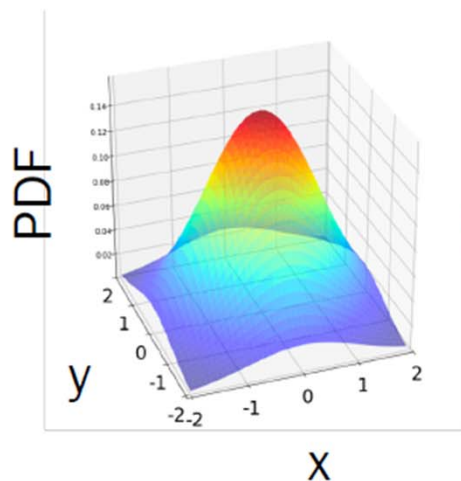
X_1 and X_2 are independent
with marginal distributions
 $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$, $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$

(X, Y) Matching (all have $\mu = (0, 0)$)

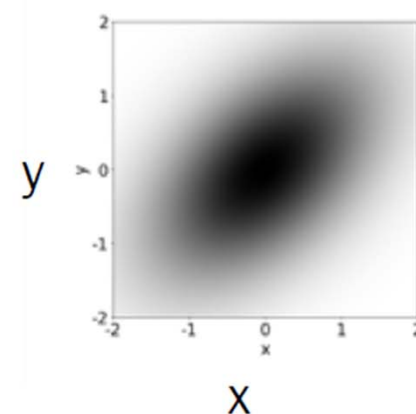
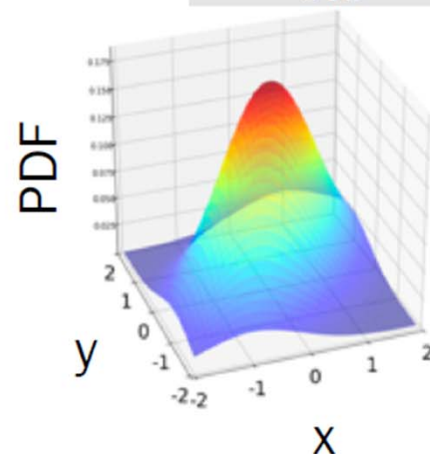


A. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
 B. $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$
 C. $\begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$
 D. $\begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix}$

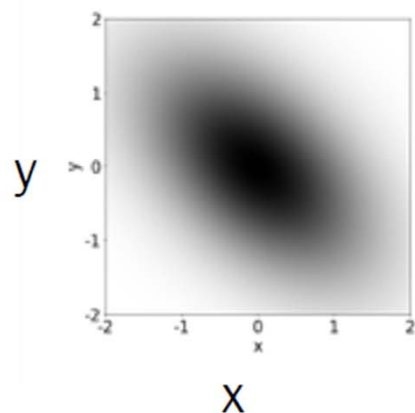
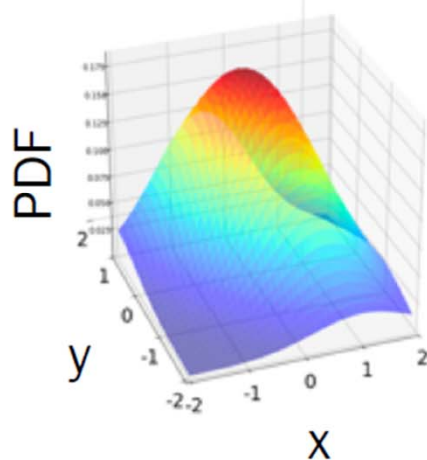
1.



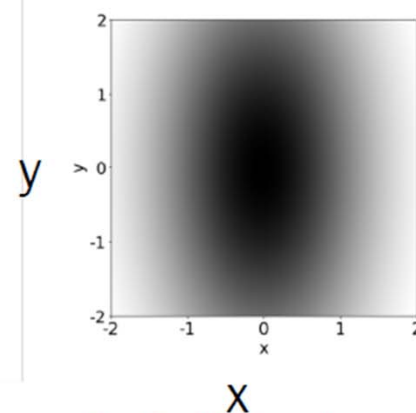
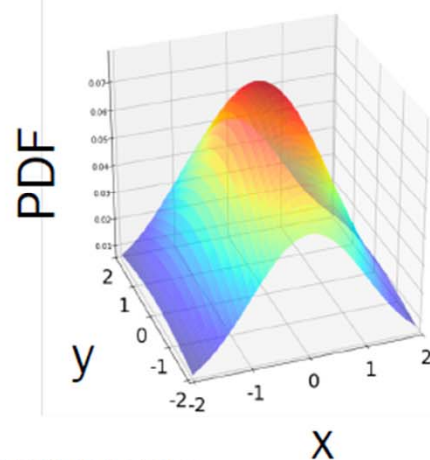
2.



3.



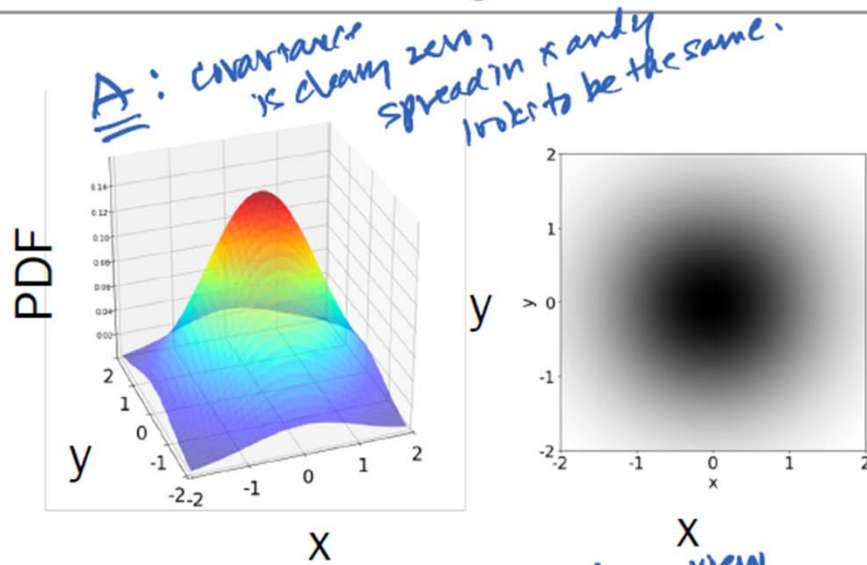
4.



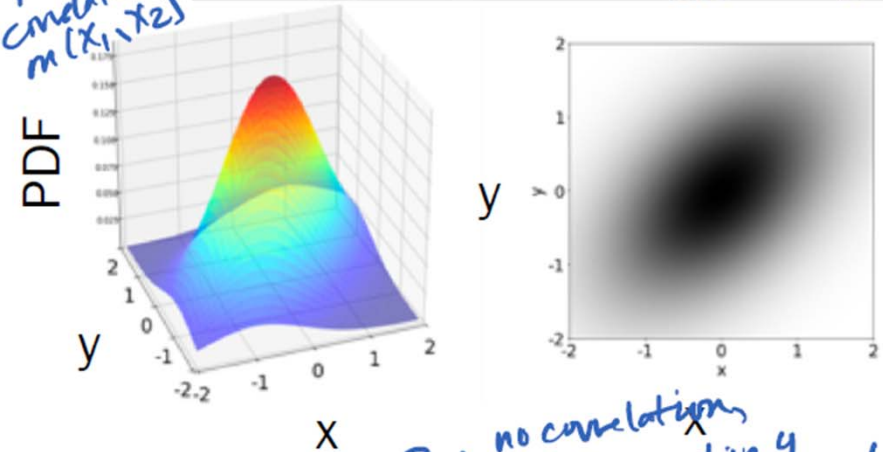
(X, Y) Matching (all have $\mu = (0, 0)$)

A. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ B. $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$
 C. $\begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$ D. $\begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix}$

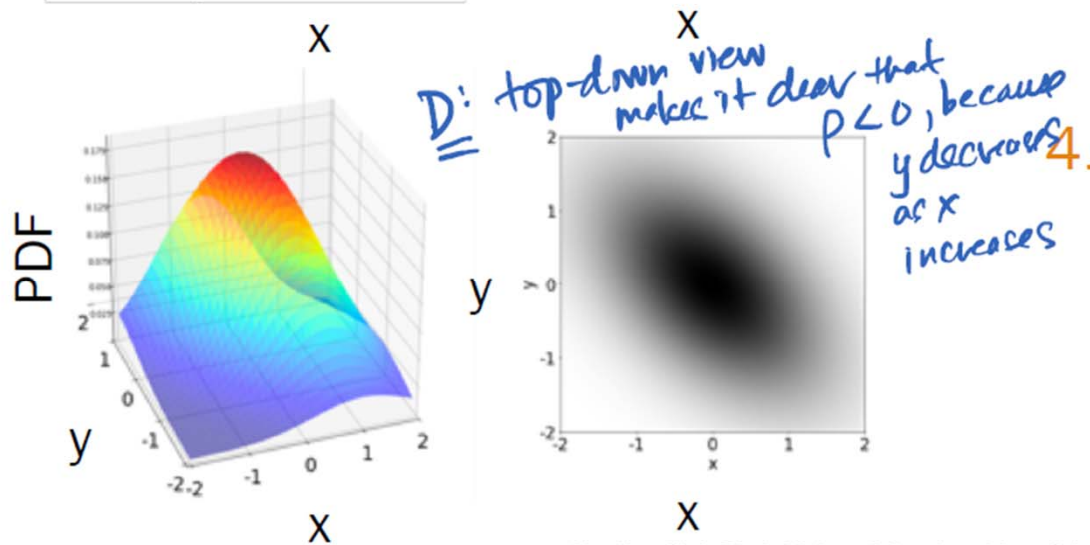
1.



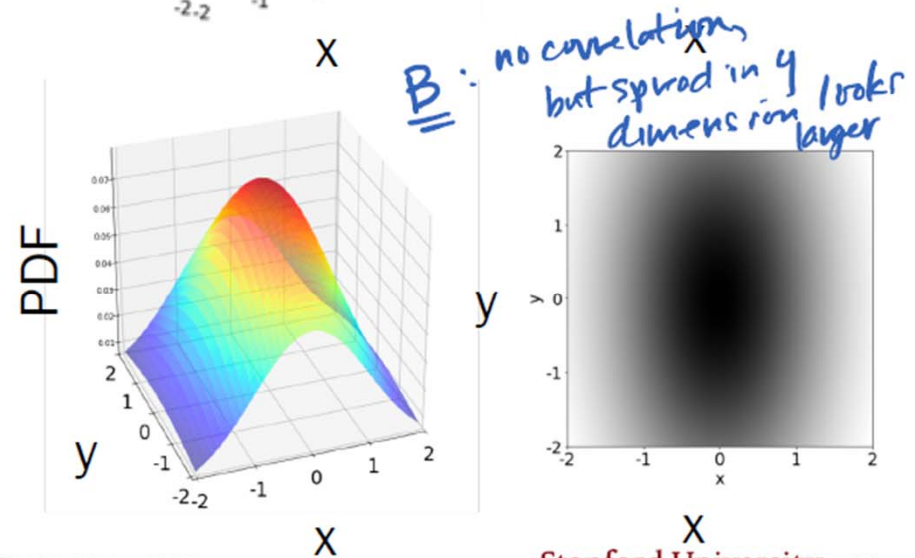
2. C: top-down view suggests positive correlation $\rho(x_1, x_2)$



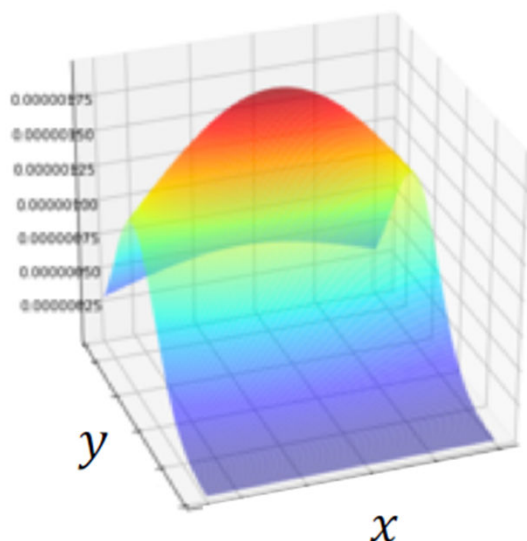
3.



4.



Why are joint PDFs useful?



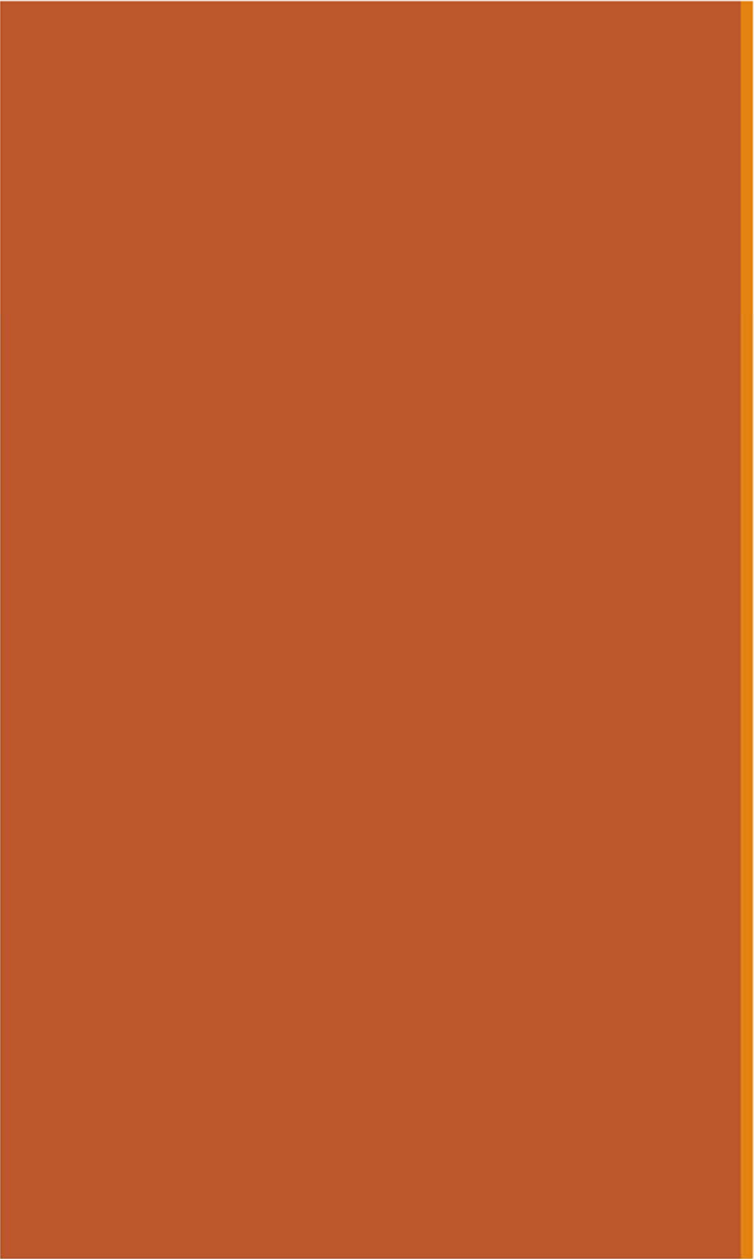
- How 2 continuous RVs vary with each other
- How continuous RV is distributed given evidence (more on Friday)
- How a continuous RV can be decomposed into 2 RVs (or vice versa)

$$P(X < Y), \\ \text{Cov}(X, Y), \rho(X, Y)$$

Given $Y = y$, the distribution of X

Distribution of $Z = X + Y$
(which is a 1-D RV!)

Independence
2-D support
Joint PDF
Joint CDF
Marginal PDF
(Friday) Conditional PDF

A solid orange horizontal bar spanning the width of the slide, positioned at the top.

Sum of Independent Gaussians

Sum of independent Gaussians

$$\Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$$

$$E[X+Y] = E[X] + E[Y] = \mu_1 + \mu_2$$
$$Var(X+Y) = Var(X) + Var(Y) = \sigma_1^2 + \sigma_2^2$$

because of independence

$$\begin{array}{l} X \sim \mathcal{N}(\mu_1, \sigma_1^2), \\ Y \sim \mathcal{N}(\mu_2, \sigma_2^2) \\ X, Y \text{ independent} \end{array} \Rightarrow X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

(proof left to [Wikipedia](#))

Holds in general case:

$$\begin{array}{l} X_i \sim \mathcal{N}(\mu_i, \sigma_i^2) \\ X_i \text{ independent for } i = 1, \dots, n \end{array} \Rightarrow \sum_{i=1}^n X_i \sim \mathcal{N}\left(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2\right)$$

Back for another playoffs game



What is the probability that the Warriors win?
How do you model zero-sum games?

$$P(A_W > A_B)$$

This is a probability of an event involving **two** random variables!

We will compute:

$$P(\underbrace{A_W - A_B}_{\text{A sum of Normals!}} > 0)$$

A sum of Normals!

Motivating idea: Zero sum games



Want: $P(\text{Warriors win}) = P(A_W - A_B > 0)$

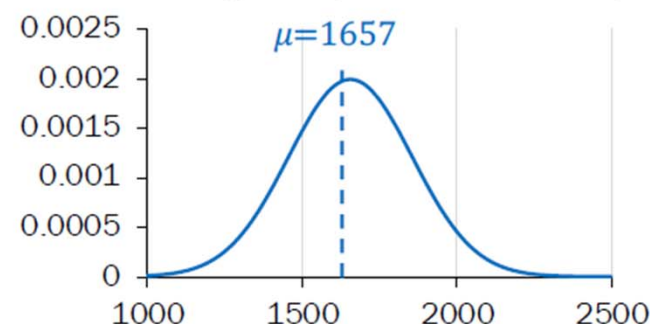
Assume A_W, A_B are independent.

Let $D = A_W - A_B$.

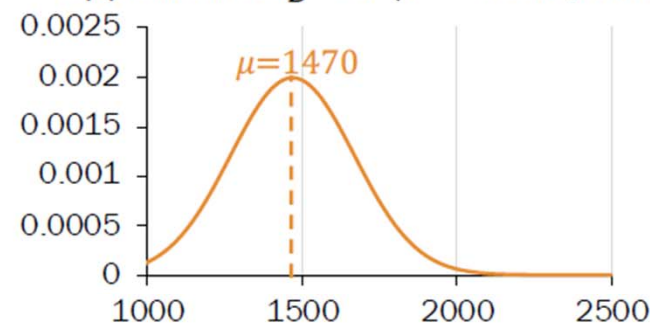
What is the distribution of D ?

- A. $D \sim \mathcal{N}(1657 - 1470, 200^2 - 200^2)$
- B. $D \sim \mathcal{N}(1657 - 1470, 200^2 + 200^2)$
- C. $D \sim \mathcal{N}(1657 + 1470, 200^2 + 200^2)$
- D. $D \sim \mathcal{N}(1657 + 1470, 200^2)$

Warriors $A_W \sim \mathcal{N}(S = 1657, 200^2)$



Opponents $A_B \sim \mathcal{N}(S = 1470, 200^2)$



Motivating idea: Zero sum games



Want: $P(\text{Warriors win}) = P(A_W - A_B > 0)$

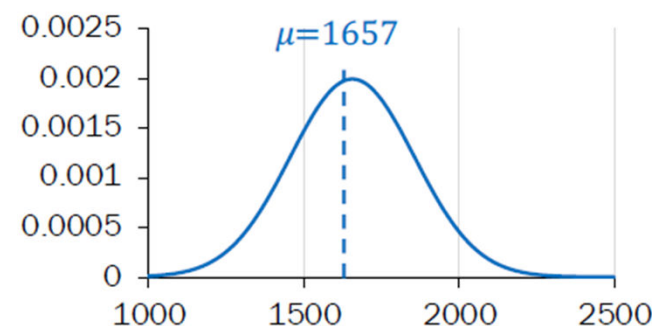
Assume A_W, A_B are independent.

Let $D = A_W - A_B$. } This is really $A_W + (-A_B)$
 $-A_B \sim \mathcal{N}(-1470, (-1)^2 200^2)$
 $\Downarrow$
 $\mathcal{N}(-1470, 200^2)$

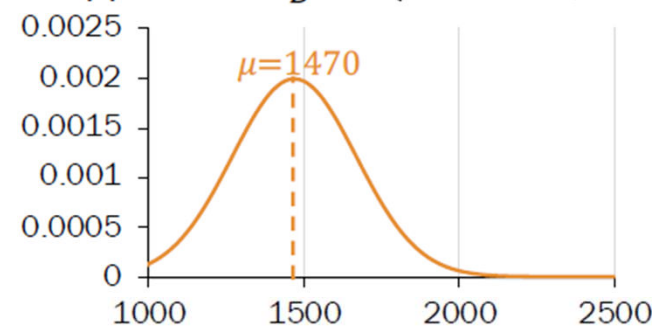
What is the distribution of D ?

- A. $D \sim \mathcal{N}(1657 - 1470, 200^2 - 200^2)$
- B. $D \sim \mathcal{N}(1657 - 1470, 200^2 + 200^2)$**
- C. $D \sim \mathcal{N}(1657 + 1470, 200^2 + 200^2)$
- D. $D \sim \mathcal{N}(1657 + 1470, 200^2)$

Warriors $A_W \sim \mathcal{N}(S = 1657, 200^2)$



Opponents $A_B \sim \mathcal{N}(S = 1470, 200^2)$



If $X \sim \mathcal{N}(\mu_1, \sigma^2)$,
then $(-X) \sim \mathcal{N}(-\mu, (-1)^2 \sigma^2 = \sigma^2)$.

Motivating idea: Zero sum games



Want: $P(\text{Warriors win}) = P(A_W - A_B > 0)$

Assume A_W, A_B are independent.

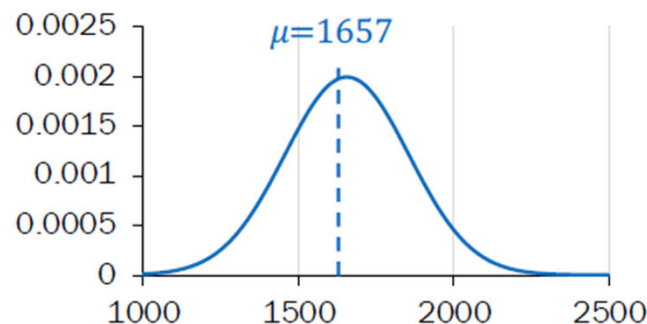
Let $D = A_W - A_B$.

$$\begin{aligned} D &\sim \mathcal{N}(1657 - 1470, 200^2 + 200^2) \\ &\sim \mathcal{N}(187, 2 \cdot 200^2) \quad \sigma \approx 282.842 \end{aligned}$$

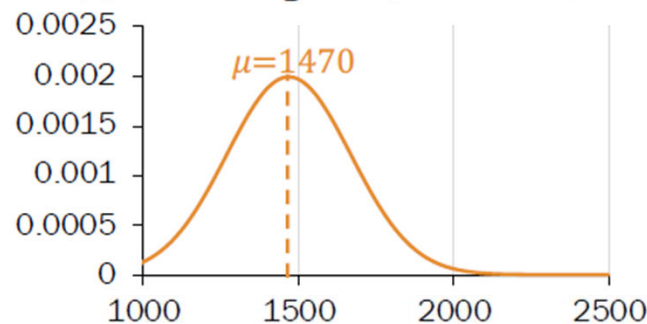
$$\begin{aligned} P(D > 0) &= 1 - F_D(0) = 1 - \Phi\left(\frac{0 - 187}{282.842}\right) \\ &\approx 0.74574 \end{aligned}$$

Compare with 0.7488, calculated by sampling!

Warriors $A_W \sim \mathcal{N}(S = 1657, 200^2)$



Opponents $A_B \sim \mathcal{N}(S = 1470, 200^2)$



```
>>> from scipy.stats import norm
>>> 1 - norm(187, 80000 ** 0.5).cdf(0)
0.7457402843526317
>>> 1 - norm(0, 1).cdf(-187 / (80000 ** 0.5))
0.7457402843526317
```

Virus infections

Suppose you are working with the WHO to initiate a response to the onset of a virus. There are two exposed groups:

- G1: 20000 people, each independently infected with $p_1 = 0.1$
- G2: 10000 people, each independently infected with $p_2 = 0.4$

What is $P(\text{people infected} \geq 6100)$? An approximation is okay.

1. Define RVs & state goal

Let $A = \#$ infected in G1.
 $A \sim \text{Bin}(20000, 0.1)$ 
 $B = \#$ infected in G2.
 $B \sim \text{Bin}(10000, 0.4)$ 

Want: $P(A + B \geq 6100)$

Strategy:

A. Sum of independent Binomials

B. Sum of independent Poissons

 C. Sum of independent Gaussians

D. Sum of independent Exponentials

no, $p_1 \neq p_2$
no, because p_1 and p_2 too large

yes!!!!

no, because mean and stdev of B are very different

Virus infections

Suppose you are working with the WHO to initiate a response to the onset of a virus. There are two exposed groups:

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& state goal

Let $A = \#$ infected in G1.
 $A \sim \text{Bin}(20000, 0.1)$
 $B = \#$ infected in G2.
 $B \sim \text{Bin}(10000, 0.4)$

2. Approximate as sum of Gaussians

$$A \approx X \sim \mathcal{N}(2000, 1800) \quad B \approx Y \sim \mathcal{N}(4000, 2400)$$

$$P(A + B \geq 6100) \approx P(X + Y \geq 6099.5) \quad \text{continuity correction}$$

3. Solve

Want: $P(A + B \geq 6100)$

Virus infections

```
>>> 1 - norm(6000, 4200 ** 0.5).cdf(6099.5)
0.06235282662988528
>>> 1 - norm(0, 1).cdf((6099.5 - 6000)/(4200 ** 0.5))
0.06235282662988528
```

Suppose you are working with the WHO to initiate a response to the onset of a virus. There are two exposed groups:

- G1: 20000 people, each independently infected with $p_1 = 0.1$
- G2: 10000 people, each independently infected with $p_2 = 0.4$

What is $P(\text{people infected} \geq 6100)$? An approximation is okay.

1. Define RVs
& state goal

Let $A = \#$ infected in G1.
 $A \sim \text{Bin}(20000, 0.1)$
 $B = \#$ infected in G2.
 $B \sim \text{Bin}(10000, 0.4)$

Want: $P(A + B \geq 6100)$

2. Approximate as sum of Gaussians

$A \approx X \sim \mathcal{N}(2000, 1800)$ $B \approx Y \sim \mathcal{N}(4000, 2400)$
 $P(A + B \geq 6100) \approx P(X + Y \geq 6099.5)$ continuity correction

3. Solve

Let $W = X + Y \sim \mathcal{N}(6000, 4200)$
 $P(W \geq 6099.5) = 1 - \Phi\left(\frac{6099.5 - 6000}{\sqrt{4200}}\right)$
 $\approx 1 - \Phi(1.53531) \approx 0.06235$

Sum of independent Gaussians

$$\begin{array}{l} X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2), \\ X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2) \\ X_1, X_2 \text{ independent} \end{array} \quad \Rightarrow \quad X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

Is this related to linear transformations of Gaussians?

Recall:

$$\text{If } Y = \underline{a}X + \underline{b}, \text{ then } Y \sim \mathcal{N}(\underline{a}\mu_X + \underline{b}, \underline{a}^2\sigma_X^2)$$

Linear transforms vs. independence



Let $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y = X + X$. What is the distribution of Y ?

- Are both approaches valid?

Independent RVs approach



Let $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$, $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$
be independent.

Then $Y = X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$

$$Y = X + X$$

$$X + X \sim \mathcal{N}(\mu + \mu, \sigma^2 + \sigma^2)?$$

$$Y \sim \mathcal{N}(\underline{2\mu}, 2\sigma^2)?$$

X is NOT
independent
of X !

Linear transform approach



Let $X \sim \mathcal{N}(\mu, \sigma^2)$.

If $Y = aX + b$,
then $Y \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$.

$$Y = 2X$$

$$Y \sim \mathcal{N}(\underline{2\mu}, 4\sigma^2)$$

For independent $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$, $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$,
 $aX_1 + bX_2 + c \sim \mathcal{N}(a\mu_1 + b\mu_2 + c, a^2\sigma_1^2 + b^2\sigma_2^2)$